

Department Overview Department Name: Department of Computer Science Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is officially accredited by the National Computing Education Accreditation Council (NCEAC). History: Founded in 2002 as one of the university's pioneer undergraduate programs during its initial inauguration. The department subsequently expanded to offer an MS degree program in 2006 and a PhD program in 2014. Core Computational Focus: Imparting foundational and technical knowledge on computer communication, networking, security, software development, database management systems, data sciences, and core programming. Career Horizons: Graduates transition into roles such as software engineers, data scientists, system administrators, and IT managers within industrial and governmental laboratories. The degree also serves as a multidisciplinary launchpad for careers in digital medicine, law, education, life sciences, and the humanities.

Vision and Objectives Vision: To provide a comprehensive computational education environment that inspires curiosity, creativity, and innovation for the enhancement of society in a positive and meaningful way. Program Educational Objectives (PEOs): Graduates are expected to:

1. Responsibly practice systematic, disciplined, and quantifiable approaches to solve complex problems across computing and allied disciplines.
2. Utilize skills and knowledge to serve industry and academia, positively impacting the socio-economic development of Pakistan.
3. Demonstrate sustained learning and adaptability to evolving technological fields through continuous professional development and self-study.
4. Bachelor of Science in Computer Science (BSCS)

Admission Requirements Academic Background: F.Sc (Pre-Engineering) / ICS with Mathematics and Physics from a recognized board (or equivalent qualification) with at least 60% marks.

Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks. Entrance Testing: Candidates must clear an entrance exam administered by NTS.

Degree Requirements Total Credit Hours: 133-134 Total Courses: 37 - 43 Minimum Academic Standing: CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory + Lab).

1st Semester • CS-110/CS-110L | Introduction to ICT & Lab | Prerequisite: None | Credit Hours: 2+1 • CS-114/CS-114L | Programming Fundamentals & Lab | Prerequisite: None | Credit Hours: 3+1 • HUM-175 | English Composition & Comprehension | Prerequisite: None | Credit Hours: 3+0 • MATHP-105 | Calculus & Analytical Geometry | Prerequisite: None | Credit Hours: 3+0 • PHY-205 / PHY-205L | Applied Physics & Lab | Prerequisite: None | Credit Hours: 2+1 • HUM-101 | Islamic Studies | Prerequisite: None | Credit Hours: 2+0

2nd Semester • CS-212/L | Object Oriented Programming & Lab | Prerequisite: Programming Fundamentals | Credit Hours: 3+1 • Hum-268 | Communication & Presentation Skills | Prerequisite: English Composition & Comprehension | Credit Hours: 3+0 • TBA | CS Supporting I - Multivariable Calculus | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0 • EE-102/L | Digital Logic & Design & Lab | Prerequisite: None | Credit Hours: 3+1 • MATHA-234 | Discrete Structures | Prerequisite: Applied Physics | Credit Hours: 3+0

3rd Semester • CS-214/L | Data Structure & Algorithms & Lab | Prerequisite: Object Oriented Programming | Credit Hours: 3+1 • CS-334 | Software Engineering | Prerequisite: None | Credit Hours: 3+0 • CE-213/L | Computer Organization & Assembly Language | Prerequisite: Digital Logic & Design | Credit Hours: 3+1 • TBA | University Elective I - Introduction to Psychology | Prerequisite: None | Credit Hours: 3+0 • MATHP-111 | Linear Algebra | Prerequisite: None | Credit Hours: 3+0

4th Semester • CS-341/L | Operating Systems & Lab | Prerequisite: Data Structure & Algorithms | Credit Hours: 3+1 • TE-307/L | Computer Networks & Lab | Prerequisite: None | Credit Hours: 3+1 • CS-321 | Theory of Automata | Prerequisite: None | Credit Hours: 3+0 • TBA | CS Supporting II- Numerical Computing | Prerequisite: None | Credit Hours: 3+0 • TBA | University Elective II - Human Resource Management | Prerequisite: None | Credit Hours: 3+0

5th Semester • HUM-266 | Technical & Business Writing | Prerequisite: None | Credit Hours: 3+0 • CS-421 | Compiler Construction | Prerequisite: Theory of Automata | Credit Hours: 3+0 • CS-412/L | Artificial Intelligence & Lab | Prerequisite: Discrete Structures | Credit Hours: 3+1 • CS-332/L | Database Systems & Lab | Prerequisite: Data Structure and Algorithm | Credit Hours: 3+1 • STAT-101 | Probability and Statistics | Prerequisite: None | Credit Hours: 3+0

6th Semester • CS-424 | Design & Analysis of Algorithm | Prerequisite: Data Structure and Algorithm | Credit Hours: 3+0 • TBA | CS Elective I - Introduction to Big Data | Prerequisite: None | Credit Hours: 2+1 • TBA | CS Elective II - Machine Learning | Prerequisite: None | Credit Hours: 2+1 • TBA | CS Supporting III - Theory of Programming Languages | Prerequisite: None | Credit Hours: 2+1 • TBA | University Elective III - Marketing | Prerequisite: None | Credit Hours: 3+0 • TBA | University Elective IV - Social Service | Prerequisite: None | Credit Hours: 0+0 (Pass/Fail)

7th Semester • TBA | Final Year Project-I (FYDP-I) | Prerequisite: Software Engineering, Technical Writing | Credit Hours: 0+3 • HUM-309 | Professional Practices | Prerequisite: None | Credit Hours: 3+0 • TBA | Parallel & Distributed Computing | Prerequisite: Operating Systems | Credit Hours: 3+0 • TBA | Information Security | Prerequisite: None | Credit Hours: 3+0 • TBA | CS Elective III - Cloud Computing | Prerequisite: None | Credit Hours: 3+0

8th Semester • TBA | Final Year Project-II (FYDP-II) | Prerequisite: Final Year Project-I | Credit Hours: 0+3 • HUM-102 | Pakistan Studies | Prerequisite: None | Credit Hours: 2+0 • TBA | CS Elective IV - Advanced DBMS | Prerequisite: None | Credit Hours: 3+0 • TBA | CS Elective V- Cryptography | Prerequisite: None | Credit Hours: 3+0 • TBA | University Elective V - Wireless Networks | Prerequisite: None | Credit Hours: 3+0

2. Master of Science (MS) in Computer Science

Admission Requirements Academic Background: 16 years of education or an equivalent qualification (such as a 4-year BS or B.E. degree) in a relevant discipline from an HEC-recognized university. Minimum Marks: At least 60% marks under the annual grading system, or a minimum CGPA of 2.5 out of 4.0. Testing: A cumulative score of at least 50% on an entry test conducted by the university or its designated authorized testing agency.

Degree Requirements Total Credit Hours: 30 Course Distribution: 8 to 10 graduate-level courses. Minimum Academic Standing: CGPA $\geq$ 2.5

3. Doctor of Philosophy (PhD) in Computer Science

Admission Requirements Academic Background: An MS, M.Phil., or equivalent 18-year master's

degree (completed after 16 years of basic schooling). The candidate's degree must feature a minimum of 30 credit hours (consisting of 24 credit hours of graduate-level coursework + 6 credit hours of research thesis) from an HEC-recognized university or DAI (Degree Awarding Institution) in a relevant field. Academic Performance: Minimum of a 1st Division pass under the annual system, or an equivalent minimum CGPA of 3.0 out of 4.0. • Testing Requirements: Prior to confirmation of program admission, candidates must successfully pass either the standard GRE (International) Subject Test with a minimum 60th percentile score, or an authorized GAT Subject Test with at least 60% marks.

Degree Requirements • Total Credit Hours: 18 • Course Distribution: 6 to 8 advanced doctoral courses, alongside mandatory dissertation work. • Minimum Academic Standing: CGPA $\geq$ 2.5